

ROV & Marine Cleanup System — Engineering Roadmap (Verbatim)

Intro

Perfect — now we turn your idea into an actual engineering execution plan.

I'll give you a clear roadmap for each team, with:

What they do
What they deliver
In what order
Dependencies between teams

Think of this like how real robotics teams operate.

■ OVERALL PROJECT PHASES (BIG PICTURE)

All teams follow this timeline:

1. Phase A: Research & Planning
2. Phase B: Design
3. Phase C: Build
4. Phase D: Integration
5. Phase E: Testing & Iteration

■ HOW TEAMS CONNECT (IMPORTANT)

Dependencies:

Mechanical → needed by everyone
Electrical → needed before control works
Control → needed before AI moves robot
AI → feeds into software
Communication → connects everything
Software → final integration

■ Recommended Build Order (CRITICAL)

1. Mechanical + Electrical (basic robot exists)
2. Control (you can move it)
3. Communication (you can command it)
4. AI (it can "see")

5. Software integration (it becomes autonomous)

6. Docking + advanced features

■ Final Advice

Don't let all teams work randomly.

■ Instead:

Set milestones

Force teams to deliver small working parts

Example:

Week 3: submarine moves manually

Week 6: stable underwater

Week 10: detects objects

■ 1■■ Mechanical Team Roadmap

■ Goal:

Build the physical structure (sub + mother ship)

Phase A:

Study:

ROV designs

Boat hull types (catamaran recommended)

Define:

Size

Weight limits

Phase B:

Design:

Submarine frame

Waterproof enclosure

Thruster mounts

Docking mechanism

Decide materials:

PVC / Acrylic / Aluminum

Phase C:

Build frame

Build waterproof housing

Install seals (O-rings, gaskets)

Phase D:

Mount:

Motors

Electronics housing

Ensure:

Balance

Neutral buoyancy

Phase E:

Water testing:

Leak test

Stability test

Fix issues

■ Deliverables:

Fully assembled waterproof submarine body

Stable mother ship hull

■ 2■■ Electrical & Power Team Roadmap

■ Goal:

Power everything safely and reliably

Phase A:

Define:

Voltage levels (5V, 12V, etc.)

Choose:

Battery type

Connectors

Phase B:

Design:

Power distribution system

ESC setup (for brushless motors)

Charging system (dock ■■ later phase)

Phase C:

Assemble:

Wiring harness

Power rails

Integrate:

Regulators

Fuses

Phase D:

Connect:

Motors

Microcontrollers

Test current draw

Phase E:

Stress testing:

Full load test

Battery life test

■ Deliverables:

Stable power system

Safe wiring (no overheating, no drops)

■ 3■■ Control Systems Team Roadmap

■ Goal:

Make the robot move correctly and stay stable

Phase A:

Learn:

PID control

Basic dynamics

Phase B:

Design:

Thruster configuration (mapping)

Define:

Movement model

Phase C:

Implement:

Motor control

Basic movement (forward, turn, up/down)

Phase D:

Add:

PID stabilization (depth + orientation)

Reaction wheel control

Phase E:

Tune:

PID parameters

Test in water

■ Deliverables:

Stable movement system

Controlled navigation

■ 4■■ AI / Computer Vision Team Roadmap

■ Goal:

Detect plastic piles and estimate size

Phase A:

Research:

Object detection

OpenCV basics

Phase B:

Build:

Simple detection (color/shape)

Collect:

Test images/videos

Phase C:

Improve:

Detect multiple objects

Cluster them into “piles”

Phase D:

Estimate:

Size / volume of pile

Decide:

Threshold (what counts as “worth collecting”)

Phase E:

Optimize:

Run on embedded system (Raspberry Pi / similar)

■ Deliverables:

Working detection system

Output: "plastic pile at (x, y) with size estimate"

■ 5■■■ Communication Team Roadmap

■ Goal:

Enable sub ↔ mother ship communication

Phase A:

Research:

Underwater communication limits

Phase B:

Choose:

Tether (recommended first)

Protocol (serial, UART, etc.)

Phase C:

Build:

Data link

Send:

Basic commands

Phase D:

Transmit:

Coordinates

Status data

Phase E:

Improve reliability:

Error handling

Signal stability

■ Deliverables:

Reliable communication system

■ 6■■■ Software Integration Team Roadmap

■ Goal:

Make everything work together

Phase A:

Define:

System architecture

Data flow

Phase B:

Build:

Command system

State machine

Example states:

Idle

Scanning

Found target

Returning

Docking

Phase C:

Integrate:

Control system

AI outputs

Phase D:

Add:

Mission logic

Autonomous behavior

Phase E:

Debug:

Full system interaction

■ Deliverables:

Fully coordinated system

“Brain” of the project